

PHY- 23 Sep - L-1

## ~~Waves & Oscillations~~

# ~~vector~~ Basics of waves & Oscillations

→ Basics of Calculus.

→ Vector analysis.

① Basic calculations of vector analysis

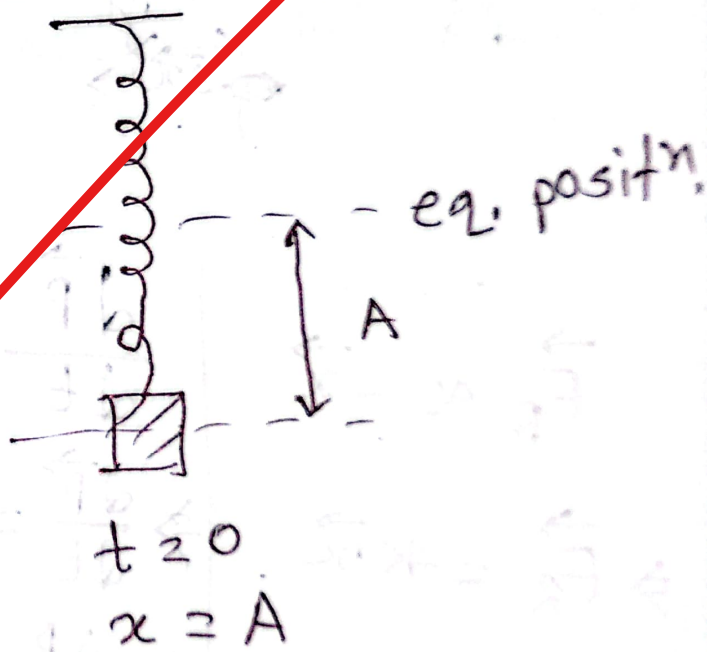
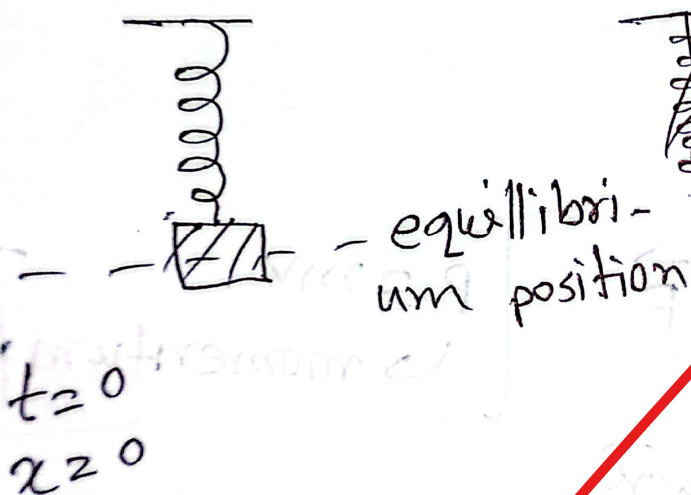
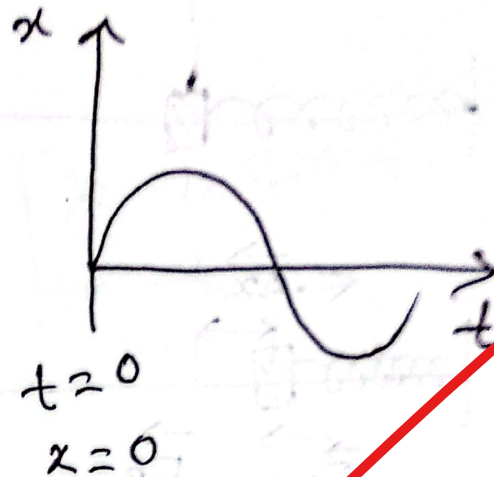
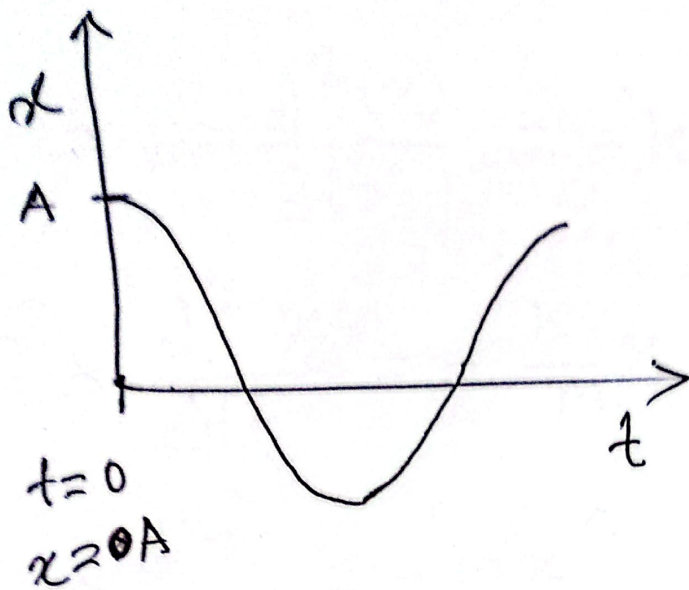
② Unit vector system.

→ Basics of motion

$$\frac{\vec{x}}{\vec{v}} = \frac{dx}{dt} \hat{i}$$

$$\begin{aligned} \frac{x}{v} &= \frac{dx}{dt} \\ a &= \frac{dv}{dt} \\ &= \frac{d^2x}{dt^2} \end{aligned}$$

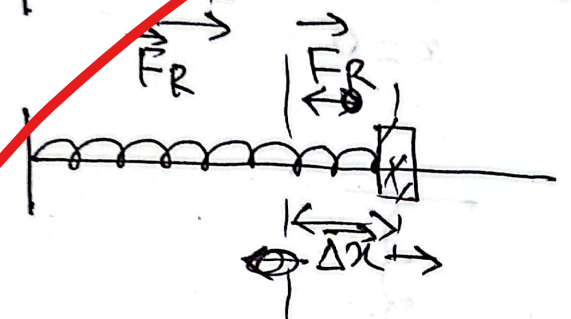
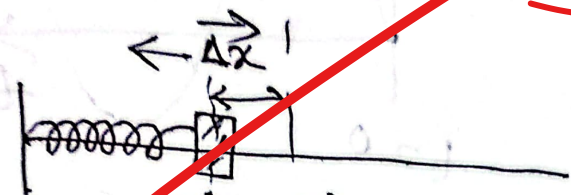
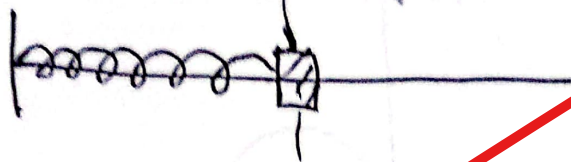
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$$T = \frac{1}{f}$$

# Mass-spring - Oscillation:

eq. posn.



$F_R$

$\frac{dv}{dt}$

$m \frac{dv}{dt}$

0.1m7d

$$\vec{F}_R \propto -\vec{x}$$

$$\Rightarrow \vec{F}_R = -k\vec{x}$$

$$\frac{d\vec{p}}{dt} = \vec{F}$$

$$p = mv$$
  
 ↳ momentum

$$\Rightarrow \frac{d\vec{p}}{dt} = -k\vec{x}$$

$$\Rightarrow \frac{dp}{dt} = -kx$$

$$\Rightarrow \frac{d(mv)}{dt} = -kx$$

$$\Rightarrow m \frac{dv}{dt} = -kx$$

✓

$$\Rightarrow m \frac{d}{dt} \left( \frac{dx}{dt} \right) = -kx$$

$$\Rightarrow m \frac{d^2(x)}{dt^2} = -kx$$

$$\Rightarrow \frac{d^2 x}{dt^2} = -\frac{k}{m} x \quad \left| \begin{array}{l} k \rightarrow \text{const.} \\ m \rightarrow \text{"} \end{array} \right.$$

$$\Rightarrow \frac{d^2 x}{dt^2} = -\omega^2 x$$

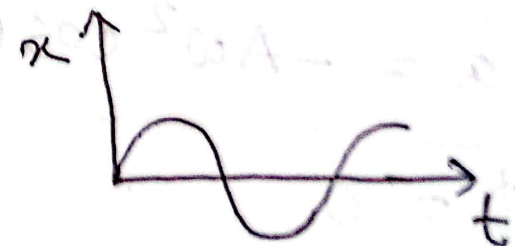
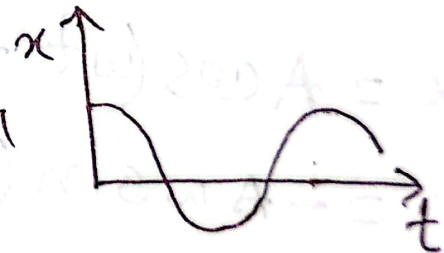
$$\Rightarrow \left[ \frac{d^2 x}{dt^2} + \omega^2 x = 0 \right] \rightarrow \text{Differential eq. of SHM.}$$

Soln:

$$\textcircled{1} x = A \cos(\omega t + \phi)$$

$$x = A \sin(\omega t + \phi)$$

Angular frequency.



$$\checkmark \boxed{x = A \cos(\omega t + \phi)} \rightarrow \text{displacement}$$

$$v = \frac{dx}{dt} = \frac{d}{dt} (A \cos(\omega t + \phi))$$

$$= -A \sin(\omega t + \phi) \cdot \frac{d}{dt} \omega t$$

$$\checkmark \boxed{v = -A \omega \sin(\omega t + \phi)} \rightarrow \text{velocity}$$

$$a = \frac{dv}{dt} = \frac{d}{dt} (-A \omega \sin(\omega t + \phi))$$

$$\checkmark \boxed{a = -A \omega^2 \cos(\omega t + \phi)} \rightarrow \text{acceleration.}$$

$$\# x = A \cos(\omega t + \phi)$$

$$v = -A \omega \sin(\omega t + \phi)$$

$$a = -A \omega^2 \cos(\omega t + \phi)$$

$$\phi = 0$$

$$x = A \cos \omega t$$

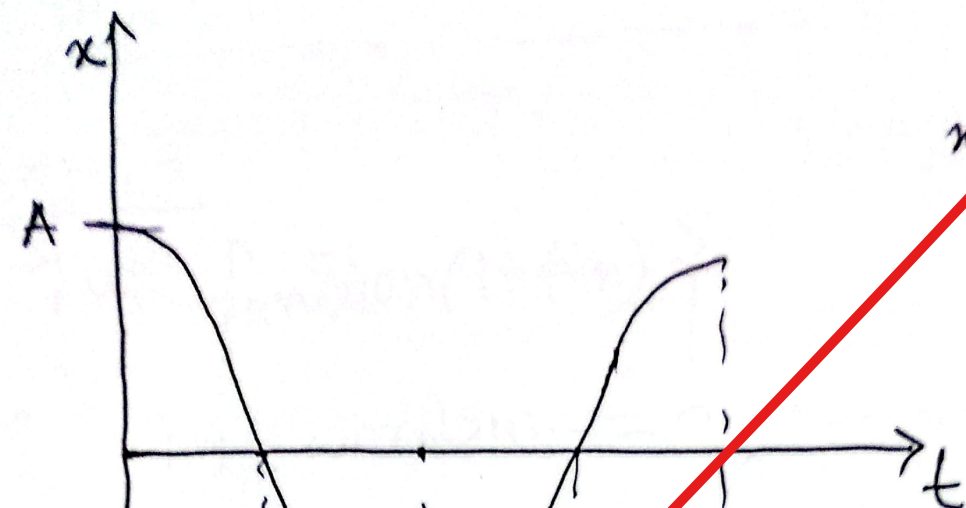
$$v = -A \omega \sin \omega t$$

$$a = -A \omega^2 \cos \omega t$$

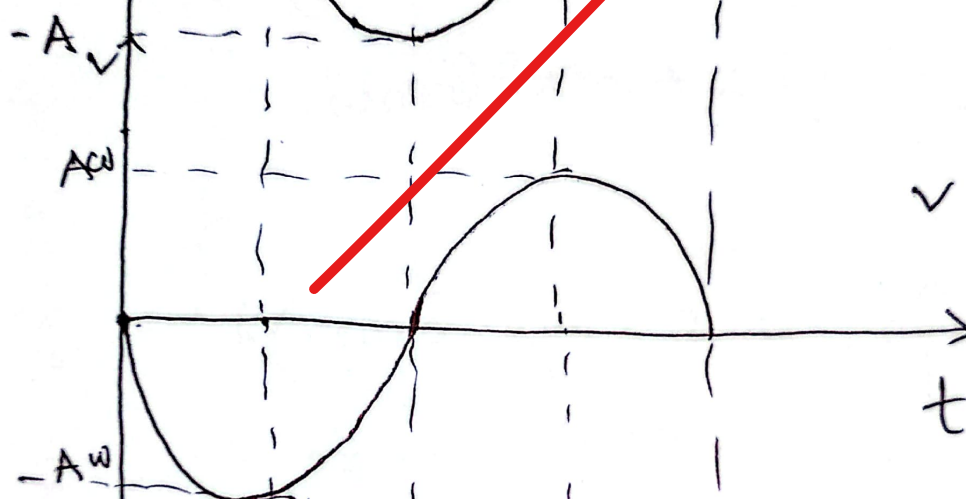
$$x_{\max} = A$$

$$v_{\max} = -A \omega$$

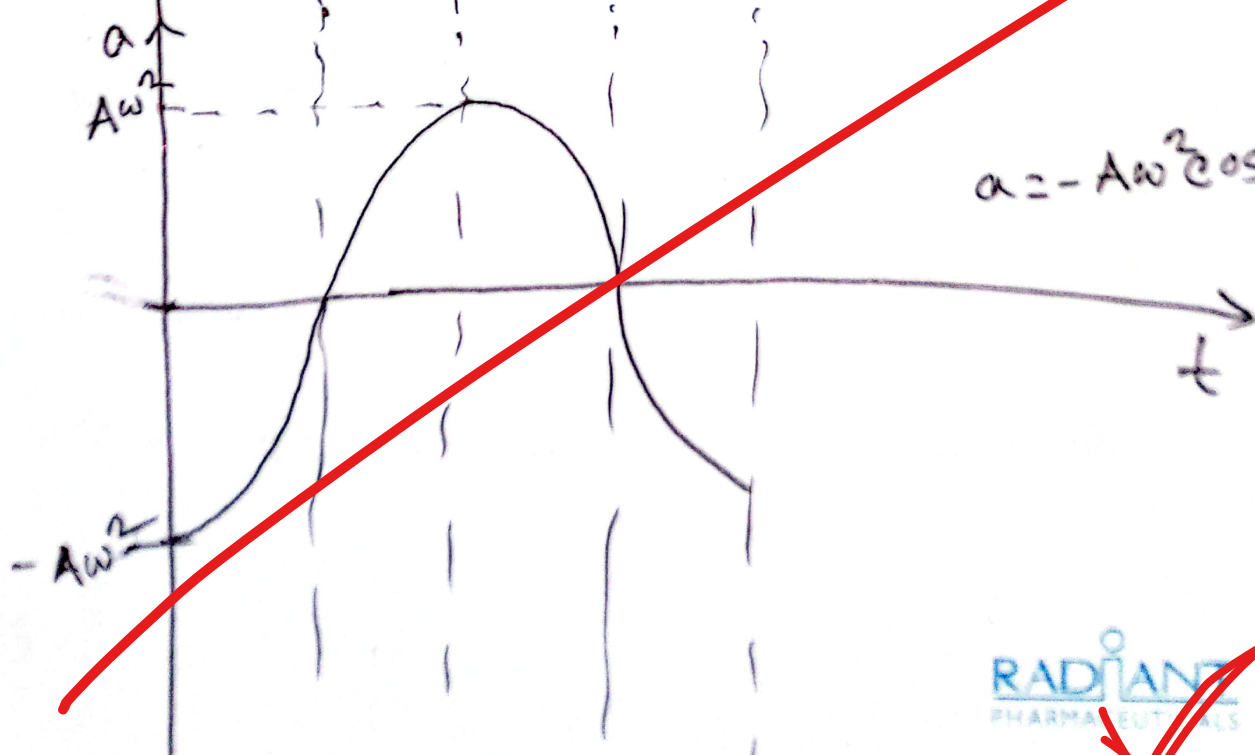
$$a_{\max} = -A \omega^2$$



$$x = A \cos \omega t$$

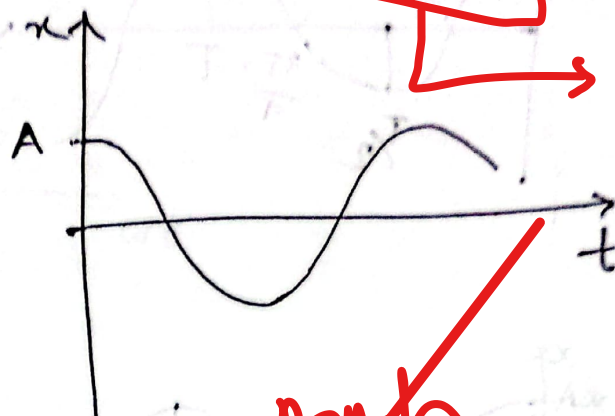
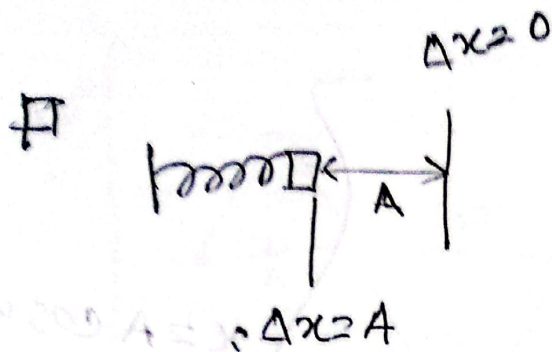


$$v = -A\omega \sin \omega t$$



$$a = -A\omega^2 \cos \omega t$$

PHY-L-3-20ct



$$x = A \cos \omega t + \phi$$

$$\phi = 0$$

$$x = A \cos \omega t$$

$$v = -A\omega \sin \omega t$$

$$a = -A\omega^2 \cos \omega t$$

$$E_p = \frac{1}{2} k A^2 \cos^2 \omega t$$

$$E_k = \frac{1}{2} k A^2 \sin^2 \omega t$$

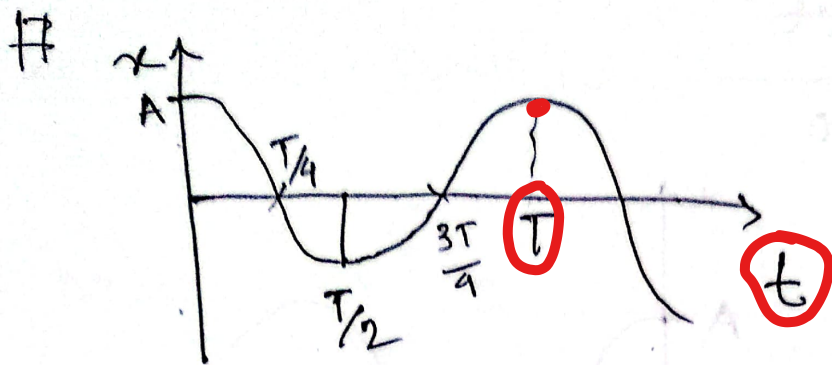
$$E_T = \frac{1}{2} k A^2$$

$$E_T = E_p + E_k = \frac{1}{2} k A^2 (\sin^2 \omega t + \cos^2 \omega t)$$

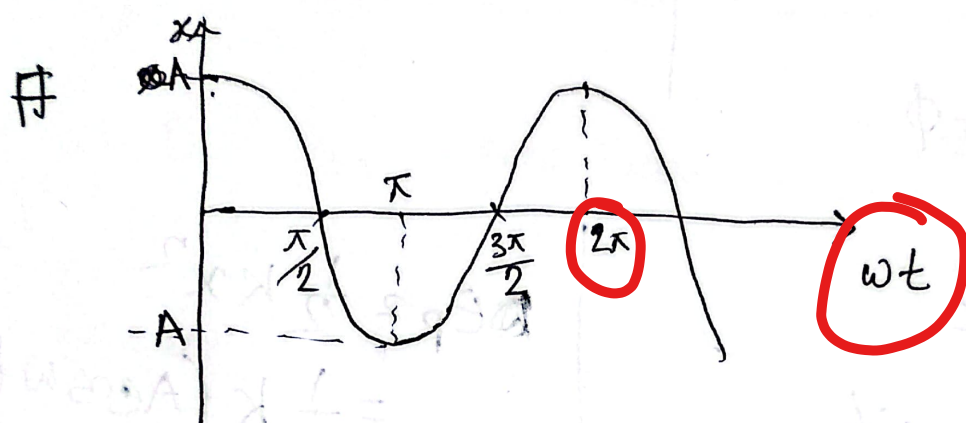
$$= \frac{1}{2} k A^2$$

$= \text{myth}$

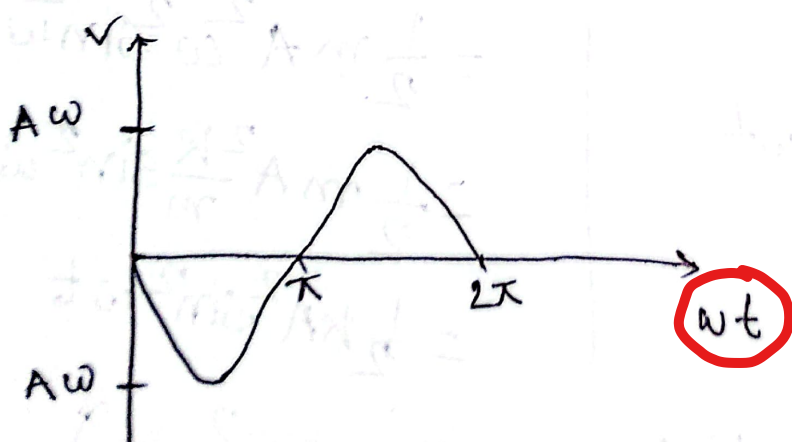
$$E_p = \frac{1}{2} k x^2$$
$$= \frac{1}{2} k (A \cos \omega t)^2$$
$$= \frac{1}{2} k A^2 \cos^2 \omega t$$
$$E_k = \frac{1}{2} m v^2$$
$$= \frac{1}{2} m A^2 \omega^2 \sin^2 \omega t$$
$$= \frac{1}{2} m A^2 \frac{k}{m} \sin^2 \omega t$$
$$= \frac{1}{2} k A^2 \sin^2 \omega t$$



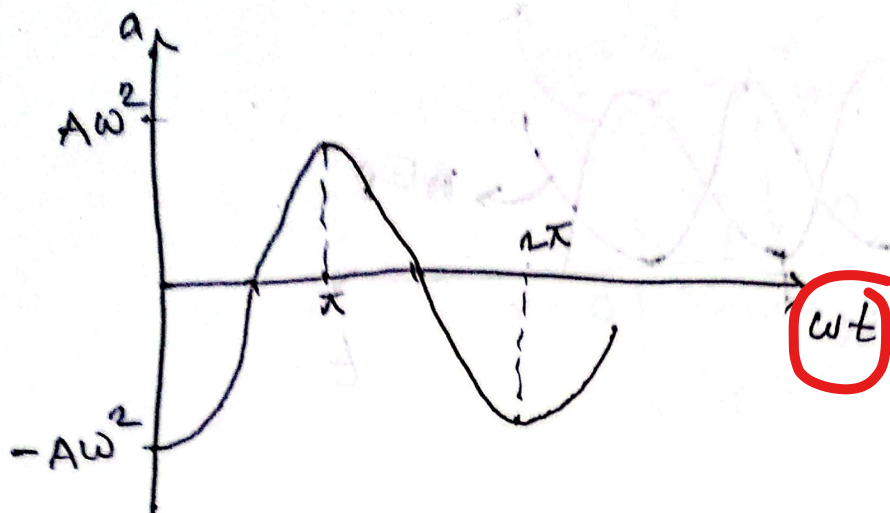
$$x = A \cos \omega t$$



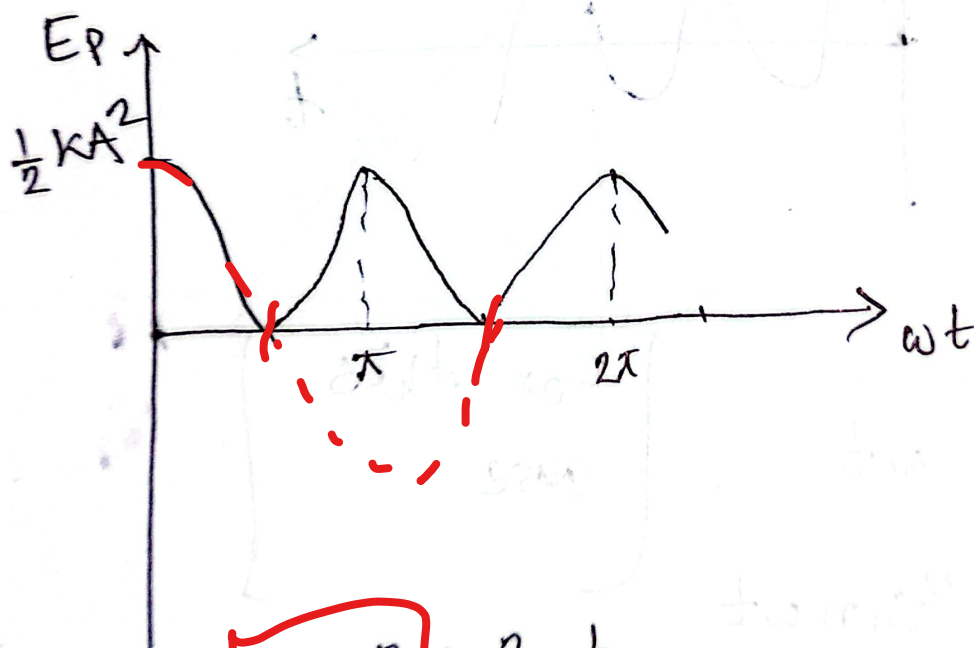
#  $v = A\omega \sin \omega t$



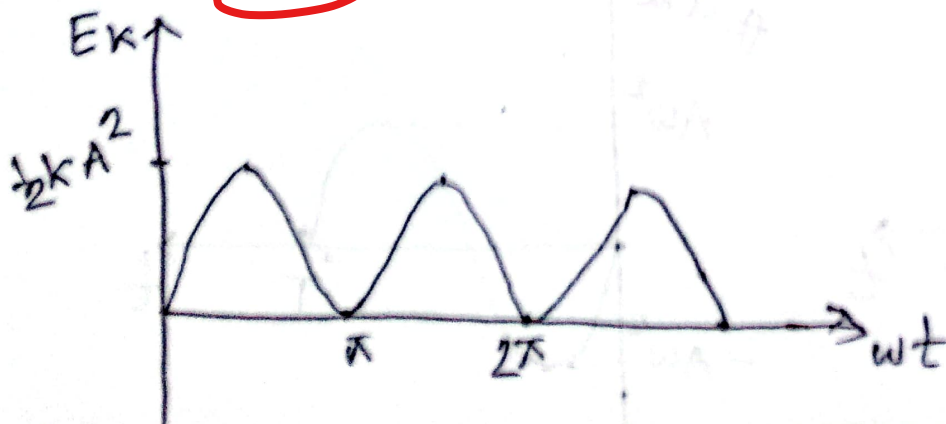
$$\# a_2 = A\omega^2 \cos \omega t$$

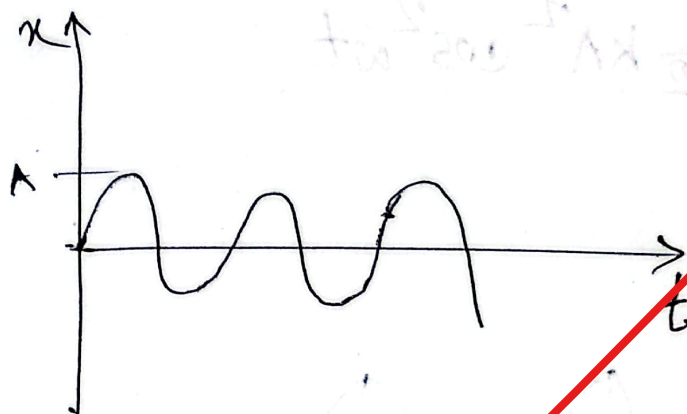
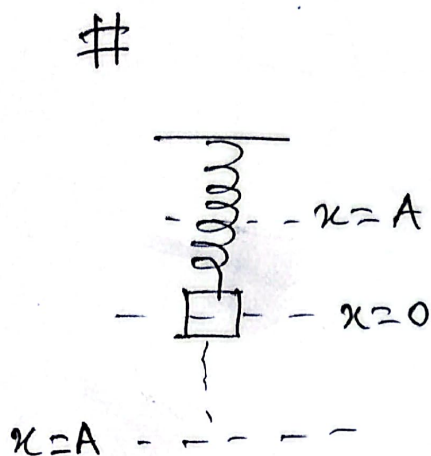
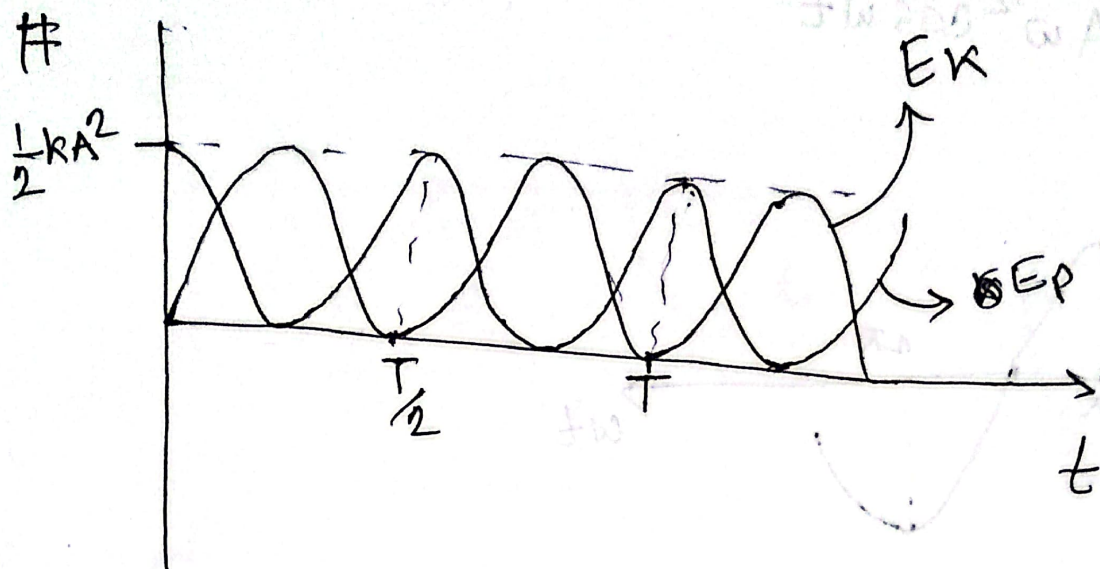


$$\# E_p = \frac{1}{2} k A^2 \cos^2 \omega t = \frac{1}{2} k x^2$$



$$\# E_k = \frac{1}{2} k A^2 \sin^2 \omega t$$



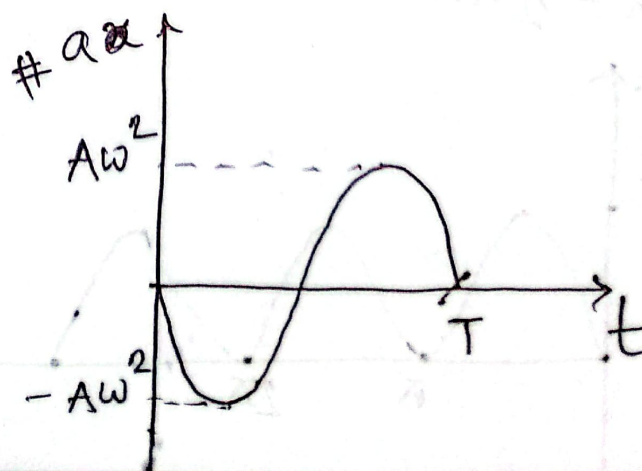
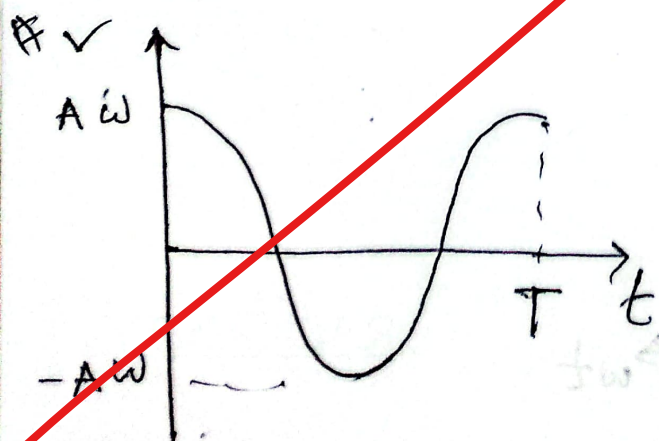


$$x = A \sin \omega t$$

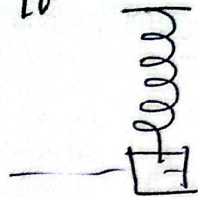
$$v = \frac{dx}{dt} = A\omega \cos \omega t$$

$$a = \frac{dv}{dt} = -A\omega^2 \sin \omega t$$

For this case

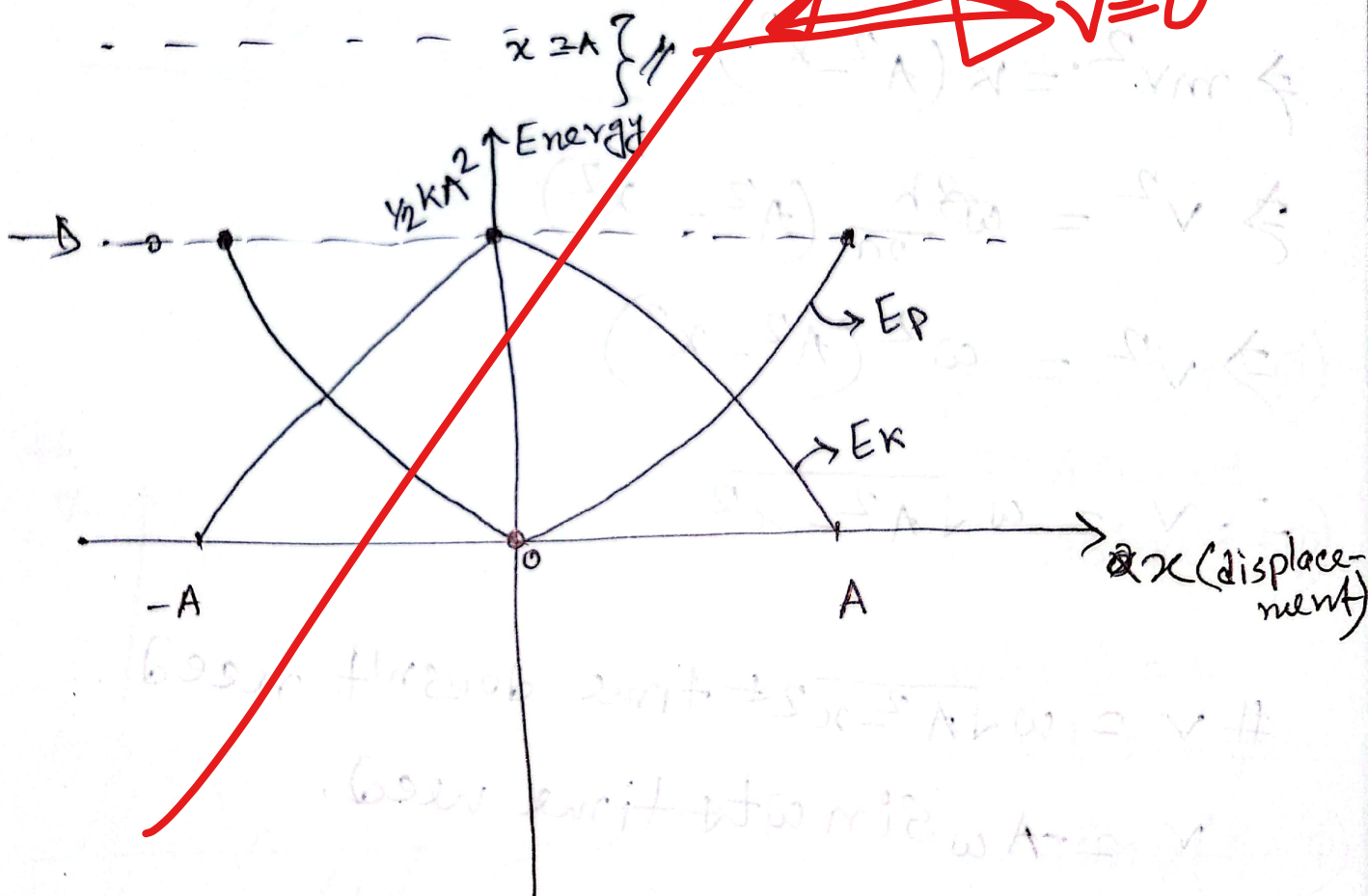


#



$$x = A \quad \left. \begin{array}{l} E_p = \frac{1}{2} k A^2 = E_T \\ E_k = 0, v = 0 \\ F_R = \text{max} \end{array} \right\}$$

$$x = 0 \quad \left\{ \begin{array}{l} F_R = 0, a = 0 \\ v_{\text{max}}, E_k = \frac{1}{2} k A^2 = E_T \\ E_p = 0 \end{array} \right.$$



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$$\rightarrow E_T = E_p + E_k$$

$$\Rightarrow \frac{1}{2} k A^2 = \frac{1}{2} k x^2 + \frac{1}{2} m v^2$$

$$\Rightarrow \frac{1}{2} m v^2 = \frac{1}{2} k A^2 - \frac{1}{2} k x^2$$

$$\Rightarrow m v^2 = k (A^2 - x^2)$$

$$\Rightarrow v^2 = \frac{k}{m} (A^2 - x^2)$$

$$\Rightarrow v^2 = \omega^2 (A^2 - x^2)$$

$$\therefore v = \omega \sqrt{A^2 - x^2}$$

~~Drop~~ ~~element~~

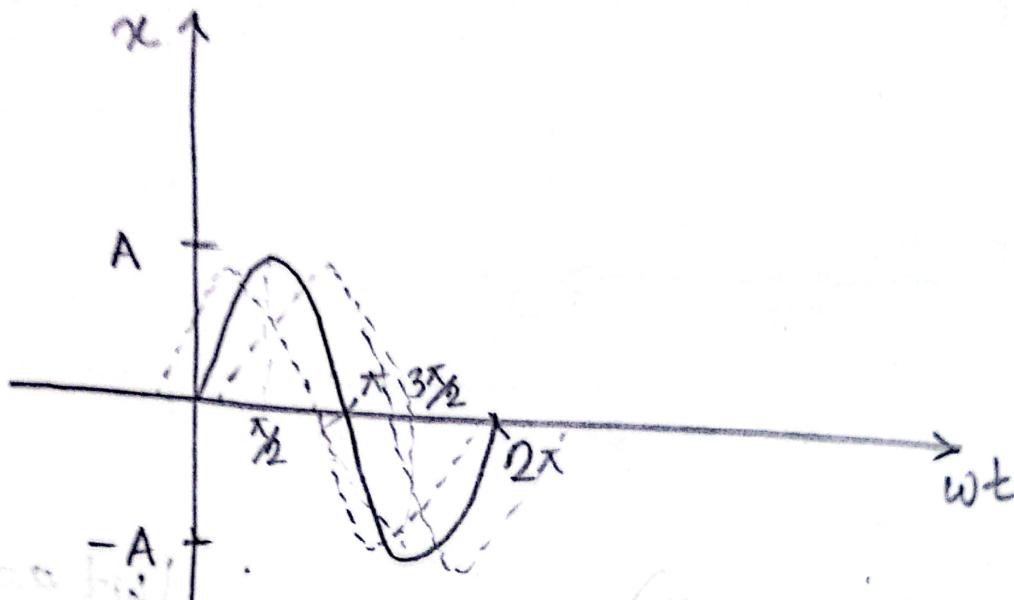
#  $v = \omega \sqrt{A^2 - x^2}$  time doesn't need

$v = -A\omega \sin \omega t$  time need.

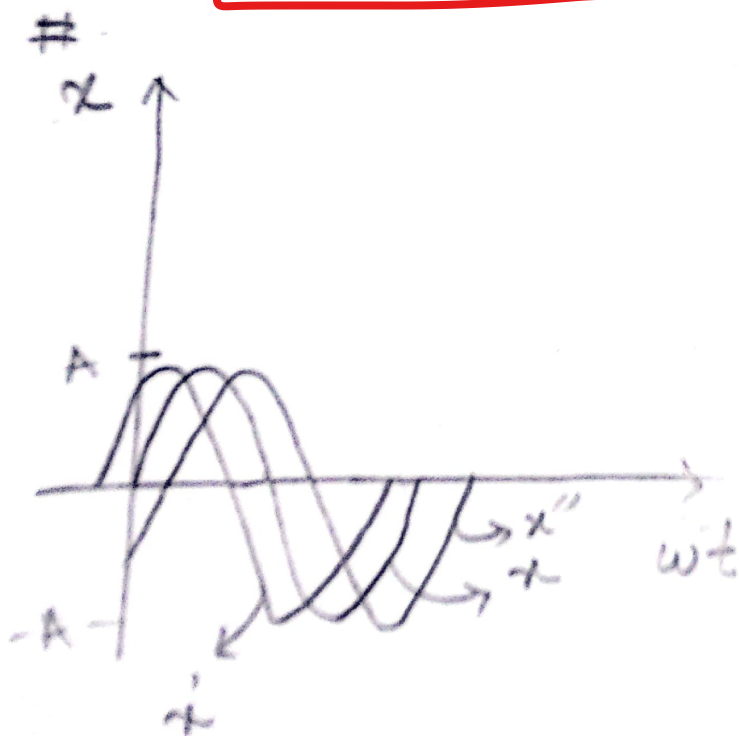


# Phase shift

$$x = A \sin \omega t$$



$$\boxed{\omega t + \phi = 0}$$



$$x'' = A \sin(\omega t - \phi)$$

$$x = A \sin \omega t$$

$$x' = A \sin(\omega t + \phi)$$

$$\omega t + \phi = 0$$

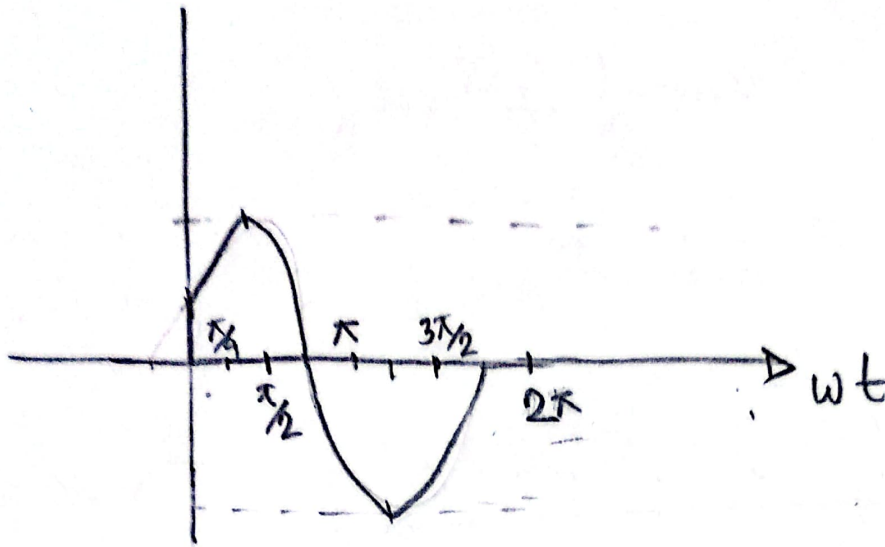
$$\therefore \omega t = -\phi$$

$$x' = A \sin \phi (-\phi + \phi)$$

$$= 0$$

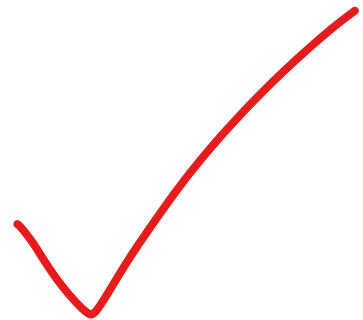
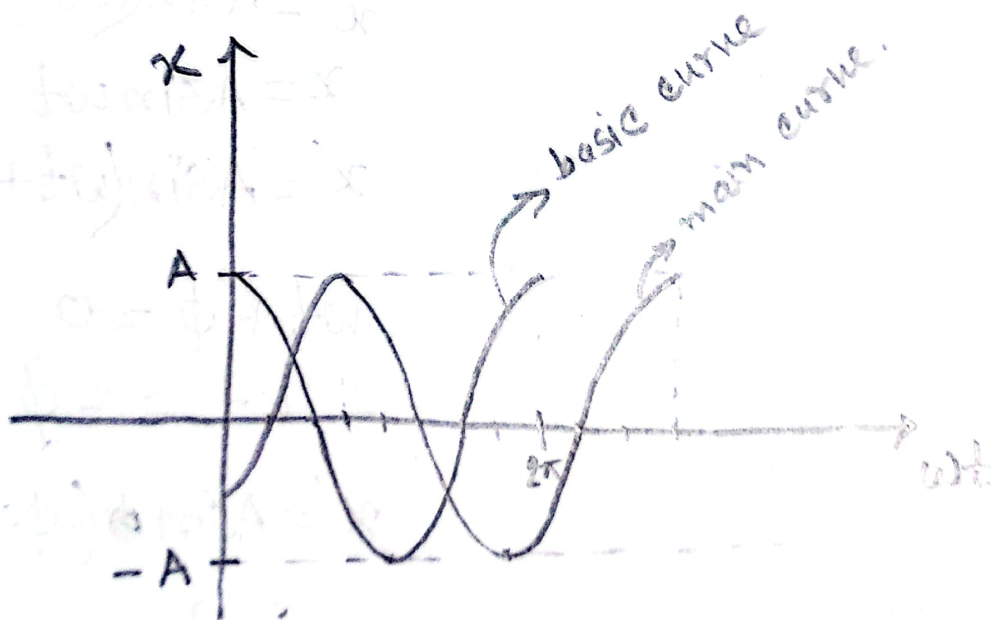
$$\# x = A \sin(\omega t + \pi/4)$$

$$\left| \begin{array}{l} \omega t + \pi/4 = 0 \\ \therefore \omega t = -\pi/4 \end{array} \right.$$



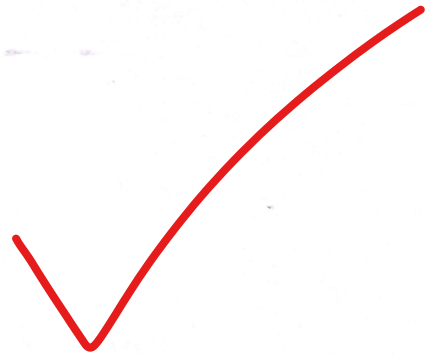
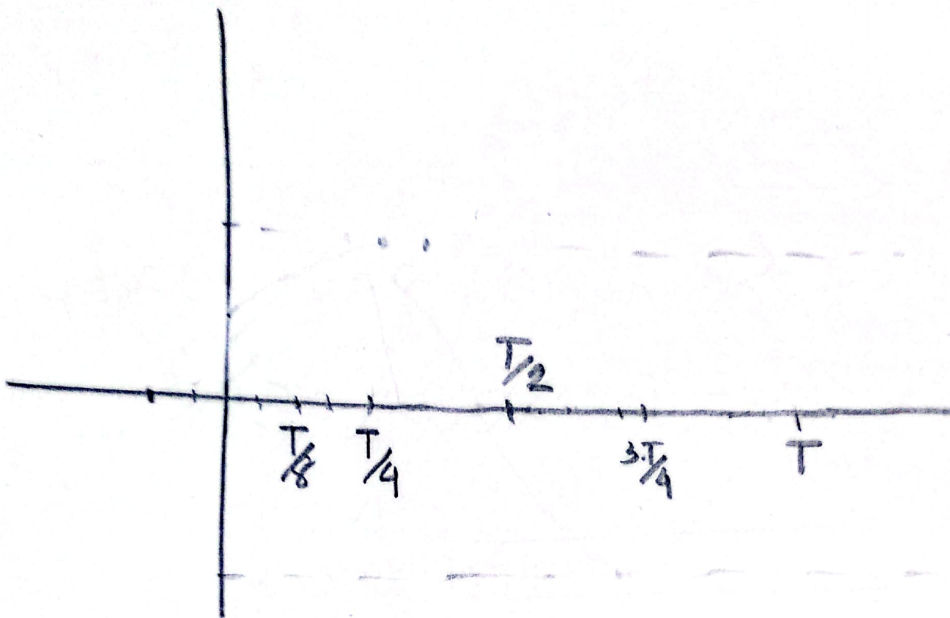
$$\# x = A \cos(\omega t - 3\pi/4)$$

$$\left| \omega t = 3\pi/4 \right.$$

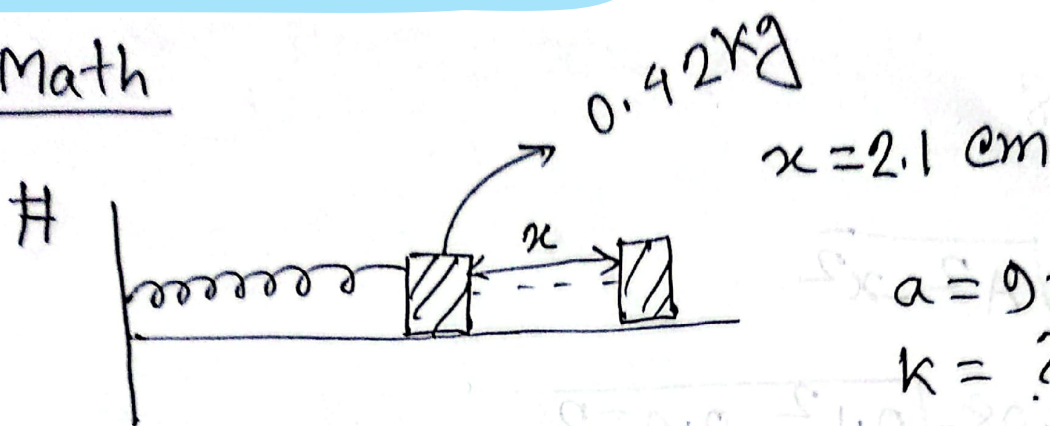


$$\# x = A \sin(\omega t + \pi/4)$$

$$\begin{aligned} \omega t &= -\pi/4 \\ \frac{2\pi}{T}t &= -\pi/4 \\ \Rightarrow t &= -\frac{T}{8} \end{aligned}$$



Math



$$\Rightarrow F = kx$$

$$\Rightarrow k = \frac{F}{x} = \frac{ma}{x} = \frac{0.42 \times 9}{2.1 \times 10^{-2}} = 180 \text{ N/m}$$

$$\# k = 10.6 \text{ N/m}, A = 0.1 \text{ m},$$

$$x = -0.1 \cos(8.08t), v = 0.808 \sin(8.08t)$$

$$a) \Rightarrow E_T = \frac{1}{2} k A^2 = 10.6 \times (0.1)^2 = \boxed{\phantom{00}}$$

$$b) \Rightarrow E_P = \frac{1}{2} k x^2 = \frac{1}{2} 10.6 (0.1 \cos(8.08t))^2 =$$

$$\Rightarrow E_K = \frac{1}{2} m v^2 = \frac{1}{2} m \{0.808 \sin(8.08t)\}^2$$

$$c) A = 0.1, x = 0.05 \text{ m}, x = -0.1 \cos 8.08 t$$

$$\Rightarrow \omega = 8.08$$

$$\therefore v = \omega \sqrt{A^2 - x^2}$$

$$= 8.08 \sqrt{0.1^2 - 0.05^2}$$

$$= 0.69$$

$$\Rightarrow E_T = \frac{1}{2} k A^2$$

$$E_P = \frac{1}{2} k \left( \frac{A}{2} \right)^2$$

$$= \frac{1}{4} \cdot \frac{1}{2} k A^2$$

$$= \frac{1}{4} E_T$$

$$E_K = E_T - E_P$$

$$= E_T - \frac{1}{4} E_T$$

$$= \frac{3}{4} E_T$$

$$\# m = 500g, A = 20 \text{ cm}, T = 0.8 \text{ sec},$$

$$v = 1 \text{ m/s}, x = ?$$

$$\Rightarrow v = \omega \sqrt{A^2 - x^2}$$

$$\Rightarrow v = \frac{2\pi}{T} \sqrt{A^2 - x^2}$$

$$\Rightarrow x = \pm \sqrt{A^2 - \frac{v^2 T^2}{4\pi^2}}$$

$$= \pm \boxed{\phantom{00}}$$

#

$$x = A \sin \omega t$$

$$v = A \omega \cos \omega t$$

$$\therefore v_{\max} = \omega A$$

$$= 2\pi f A$$

#  $k = 196 \text{ N/m}$ ,  $m = 2 \text{ kg}$

a)  $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{196}{2}} = 7\sqrt{2}$

$\Rightarrow \omega = 2\pi f = \frac{2\pi}{T}$

$\Rightarrow T = \frac{2\pi}{\omega}$

b)  $x = 5 \cos 7\sqrt{2} t$

#  $A = 20 \text{ cm}$

$\Rightarrow T = \frac{10}{15} = 0.67 \text{ sec}$

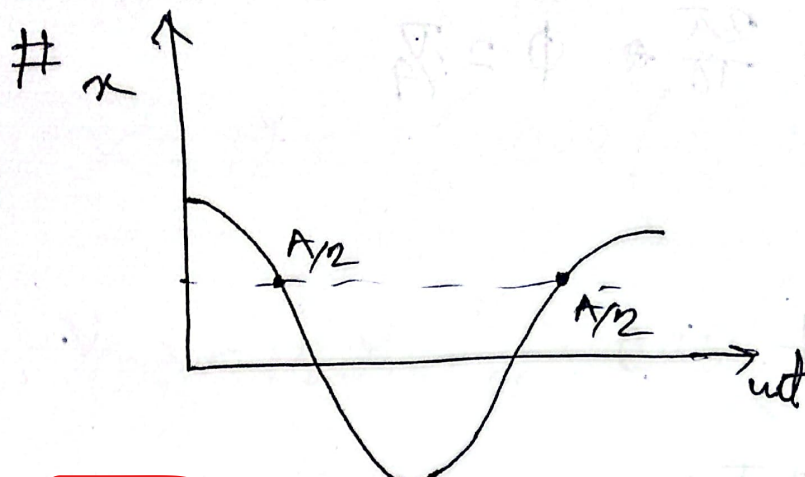
$\therefore v_{\max} = \omega A = \frac{2\pi}{T} A = \boxed{\phantom{000}} \text{ m/s}$

$\therefore x = 0.2 \cos\left(\frac{2\pi}{T} \cdot t\right) = 0.2 \cos(9.38t)$

$\therefore x = 0.2 \cos(9.38 \times 0.8) = \boxed{\phantom{000}}$

$\therefore v = \omega \sqrt{A^2 - x^2} = \boxed{\phantom{000}}$

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$$x = A \cos \omega t$$

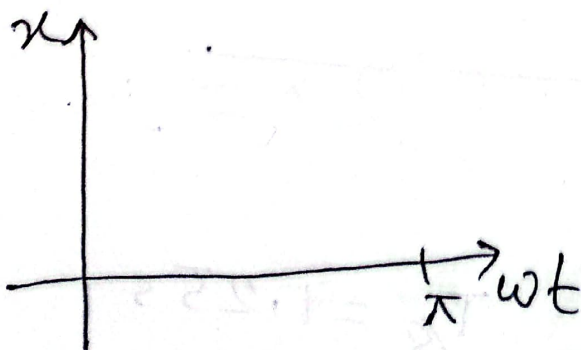
$$\frac{A}{2} = A \cos \omega t$$

$$\Rightarrow \frac{1}{2} = \cos \omega t$$

$$\Rightarrow t = \frac{1}{\omega} \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{T}{2\pi} \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{T}{2\pi} \times \frac{\pi}{3} = \frac{T}{6}$$



$$\# y = 12 \sin\left(\frac{2\pi t}{10} + \frac{\pi}{4}\right)$$

$$A = 12, \quad \omega = \frac{2\pi}{10} \quad \phi = \frac{\pi}{4}$$

Soln:

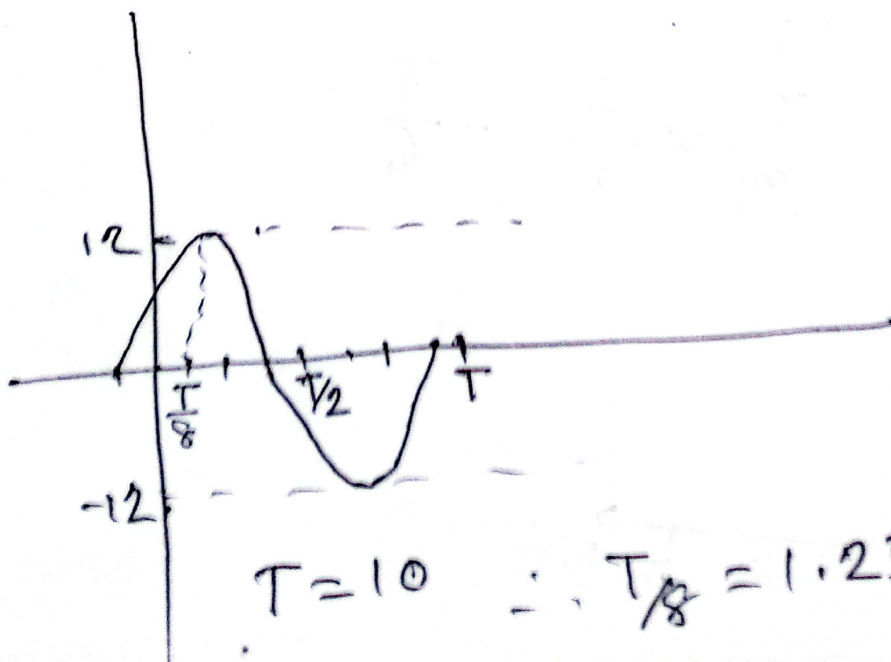
$$i) A = 12 \text{ cm}$$

$$ii) \omega = 2\pi f = \frac{2\pi}{10}$$

$$\therefore f = \frac{1}{10} = 0.1 \text{ Hz}$$

$$iii) y(1.25) = 12 \sin\left(\frac{2\pi \times 1.25}{10} + \frac{\pi}{4}\right)$$

$$= 12$$



$$T = 10 \quad \therefore T/8 = 1.25 \text{ s}$$

iv)

$$y = 12 \sin\left(\frac{2\pi}{10}t + \frac{\pi}{4}\right)$$

$$v = 12 \times \frac{2\pi}{10} \cos\left(\frac{2\pi}{10}t + \frac{\pi}{4}\right)$$

$$\therefore v_{(2.5)} = 12 \times \frac{2\pi}{10} \cos\left(\frac{2\pi}{10} \times 2.5 + \frac{\pi}{4}\right)$$

$$= \frac{12\pi}{5} \cos\left(\frac{\pi}{2} + \frac{\pi}{4}\right)$$

$$= -5.33 \text{ m/s}$$

$$v) \quad v = \frac{12\pi}{5} \cos\left(\frac{2\pi}{10}t + \frac{\pi}{4}\right)$$

$$a = -\frac{12\pi^2}{25} \sin\left(\frac{2\pi}{10}t + \frac{\pi}{4}\right)$$

$$a_{(5)} = -\frac{12\pi^2}{25} \sin\left(\frac{2\pi}{10} \times 5 + \frac{\pi}{4}\right)$$

$$= -\frac{12\pi^2}{25} \sin \frac{\pi}{4}$$

$$= -3.35 \text{ m/s}^2$$

$$\# y = 10 \sin(10t - \pi/6)$$

$$i) \omega = 2\pi f = 10$$

$$\therefore f = 5/\pi = 1.59 \text{ Hz}$$

$$ii) T = \frac{\pi}{5} = 0.628 \text{ s}$$

$$iii) \cancel{A = 10 \text{ cm}} \quad y_{\max} = A = 10$$

$$iv) v_{\max} = \omega A = 10 \times 10 = 100$$

$$v) a_{\max} = -A\omega^2 = -10 \times 10^2 = -1000 \text{ m/s}^2$$

#  $t = 5 \text{ sec}$ , what is K.E?

$$y = 10 \sin(10t - \pi/6), \quad k = 19.6 \text{ N/m}$$

Sol<sup>n</sup>: M-1

$$y_5 = -7.09 \text{ m.}$$

$$E_p = \frac{1}{2} k y^2$$

$$= \frac{1}{2} \times 19.6 \times (-7.09)^2$$

$$= 493.6 \text{ J}$$

$$E_T = \frac{1}{2} k A^2$$

$$= \frac{1}{2} \times 19.6 \times 100$$

$$= 983 \text{ J}$$

$$\therefore E_k = E_T - E_p$$

$$= 489.4 \text{ J}$$

M-2

$$y = 10 \sin(10t - \pi/6)$$

$$v = 100 \cos(10t - \pi/6)$$

$$\therefore v(5) = 100 \cos(10 \times 5 - \pi/6) \\ = 70.45 \text{ m/s}$$

$$E_k = \frac{1}{2} m v^2 \quad \cancel{k} = \omega^2 = \frac{k}{m} \\ = \frac{1}{2} \times \frac{k}{\omega^2} (70.45)^2 \quad \therefore m = \frac{k}{\omega^2} \\ = 480.6 \text{ g}$$

$$\# y = 10 \sin(10t - \pi/6), k = 19.6 \text{ N/m}$$

Soln:  $v = 100 \cos(10t - \pi/6)$

$$E_p = \frac{1}{2} k y^2 = \frac{1}{2} k 100 \sin^2(10t - \pi/6) \\ = \frac{1}{2} 19.6 \times 100 \sin^2(10t - \pi/6)$$

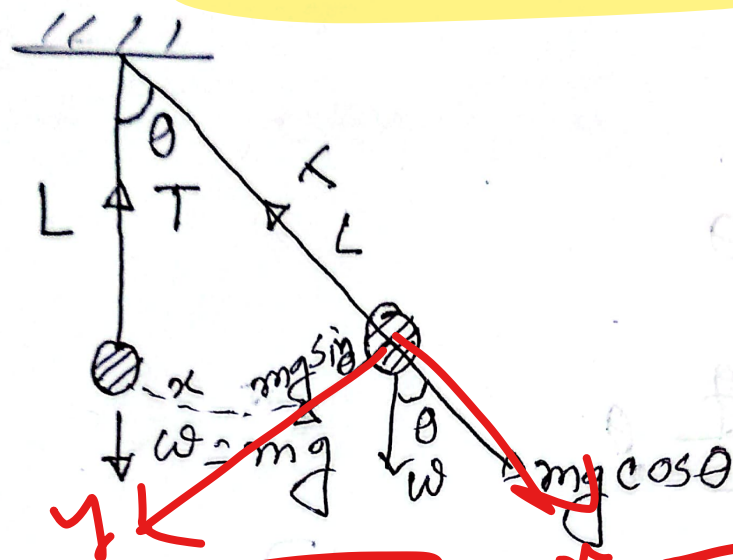
$$E_p = 0.80 \sin^2(10t - \pi/6)$$

$$\therefore E_p(5) = 0.80 \sin^2(10 \times 5 - \pi/6)$$

$$= 403.67$$

PHY-L-7-14 oct

## Simple Pendulum



$T \rightarrow$  Tension  
 $w \rightarrow$  weight

#

$$T = mg \cos \theta$$

#

$$F = -mg \sin \theta$$

$$F = -mg \theta$$

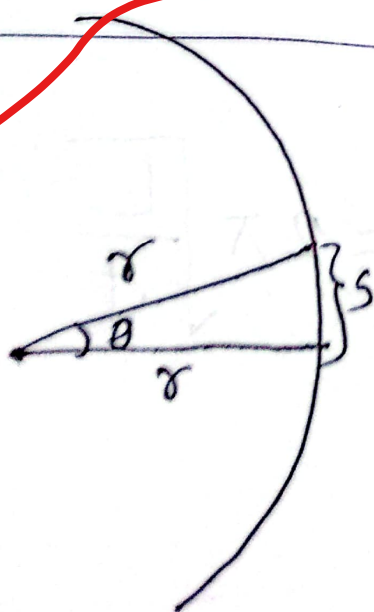
$$x = L \theta$$

if,  $\theta$  is very small,  
 $\sin \theta \approx \theta$

$$\Rightarrow \frac{d^2 x}{dt^2} = L \frac{d^2 \theta}{dt^2}$$

$$s = r \theta$$

$$\Rightarrow a = L \frac{d^2 \theta}{dt^2}$$



$$\therefore F = -mg \sin \theta \rightarrow$$

Taking to its normal position

$$\Rightarrow ma = -mg \theta$$

$$\Rightarrow a = -g \theta$$

$$\Rightarrow L \frac{d^2 \theta}{dt^2} = -g \theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} = -\frac{g}{L} \theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} = -\omega^2 \theta \quad \left[ \omega^2 = \frac{g}{L} \right]$$

$$\boxed{\frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0} \rightarrow \text{SHM}$$

$$\# \omega^2 = \frac{g}{L}$$

$$\Rightarrow \omega = \sqrt{\frac{g}{L}}$$

$$\Rightarrow \frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

$$\therefore T = 2\pi \sqrt{\frac{L}{g}}$$



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$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$\# T_1 = 2s, \quad g_1 = 9.78 \text{ m/s}^2$$

$$T_2 = 3.5s, \quad g_2 = ?$$

$$\rightarrow T_1 = 2\pi \sqrt{\frac{L}{g_1}} \text{ (i)}, \quad T_2 = 2\pi \sqrt{\frac{L}{g_2}} \text{ (ii)}$$

$$(i) \div (ii) \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{g_2}{g_1}}$$

$$\Rightarrow g_2/g_1 = \frac{T_1^2}{T_2^2}$$

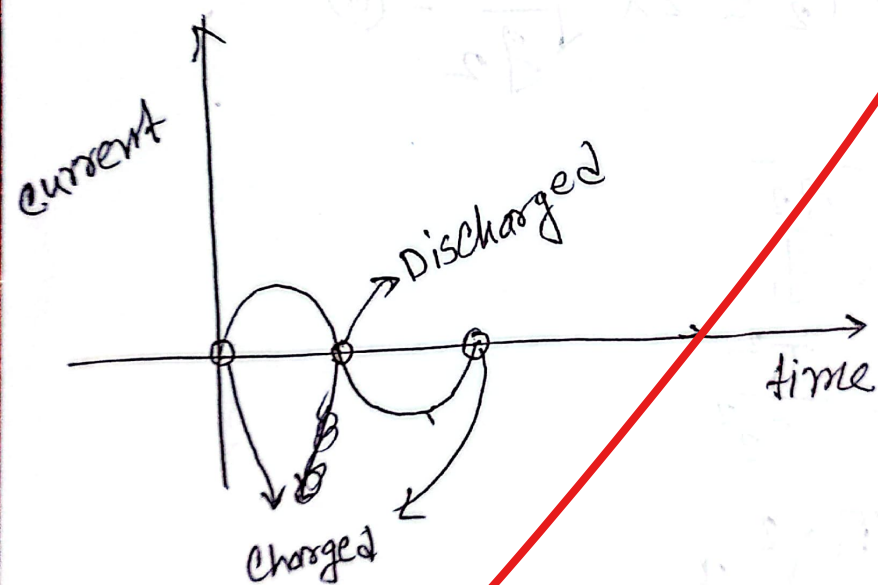
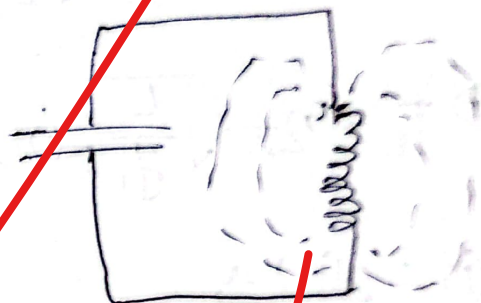
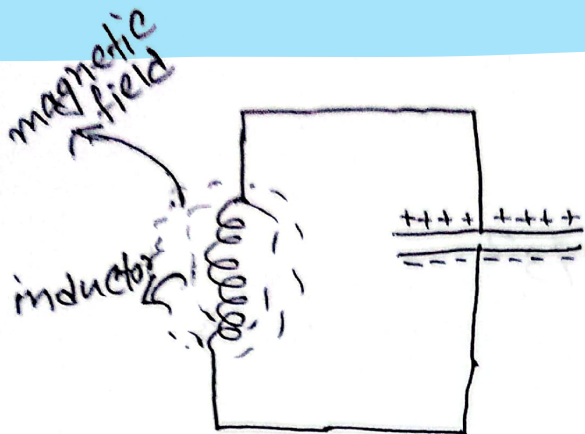
$$\Rightarrow g_2 = \left(\frac{T_1}{T_2}\right)^2 g_1$$

$$\# T_1 = T, \quad L_1 = L, \quad L_2 = 2L, \quad T_2 = ?$$

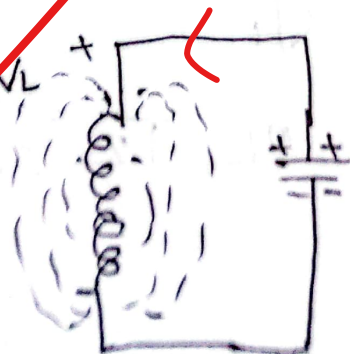
$$\Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$$

$$\Rightarrow T_2 = \sqrt{\frac{L_2}{L_1}} \times T_1 = \sqrt{\frac{2L}{L}} \times T = \sqrt{2} T$$

# LC-Circuit



$$V_L = L \frac{di}{dt}$$



$$Q = CV$$

$$V = \frac{Q}{C}$$

$$V_C = \frac{Q}{C}$$

$$V_L = L \frac{di}{dt}$$

$L \rightarrow$  inductance

$i \rightarrow$  current flow

Proof

$$V_L + V_C = 0 \quad [\text{Karchof's voltage Law}]$$

$$\Rightarrow L \frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d}{dt} \left( \frac{dQ}{dt} \right) + \frac{Q}{C} = 0$$

$$i = \frac{dQ}{dt}$$

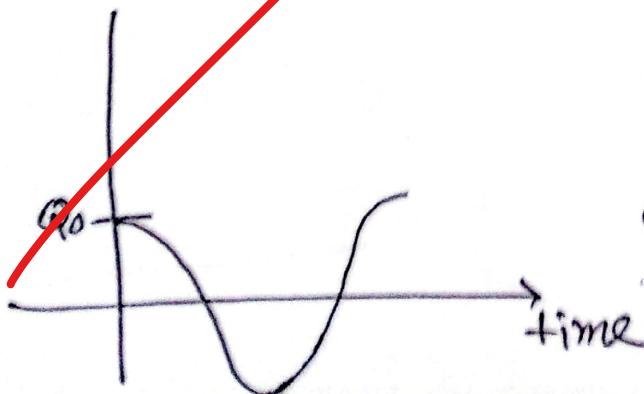
$$\Rightarrow \frac{d^2 Q}{dt^2} + \frac{1}{LC} Q = 0$$

$$\Rightarrow \frac{d^2 Q}{dt^2} + \omega^2 Q = 0$$

$$\omega^2 = \frac{1}{LC}$$

$$Q = Q_0 \sin \omega t$$

$$Q = Q_0 \cos \omega t \quad [\text{most suitable}]$$



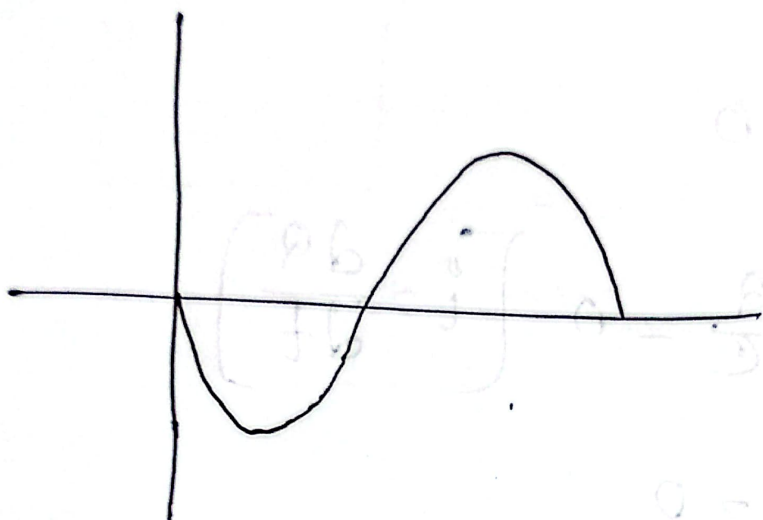
cosine

$$I = \frac{dQ}{dt} = \frac{d}{dt} Q_0 \cos \omega t$$

$$= -\omega Q_0 \sin \omega t$$

$$\Rightarrow I = -I_0 \sin \omega t$$

$$I_0 = \omega Q_0$$



$$\# \omega^2 = \frac{1}{LC}$$

$$\Rightarrow \frac{2\pi}{T} = \frac{1}{\sqrt{LC}}$$

$$\therefore T = 2\pi \sqrt{LC}$$

$$f = \frac{1}{2\pi \sqrt{LC}}$$

$$Q = Q_0 \cos \omega t$$

$$I = -I_0 \sin \omega t \quad [I_0 = \omega Q_0]$$

and